

Highly Dynamic Precision Dosing with Piezo Technology

PICMA® Actuators for Industrial Dispensing

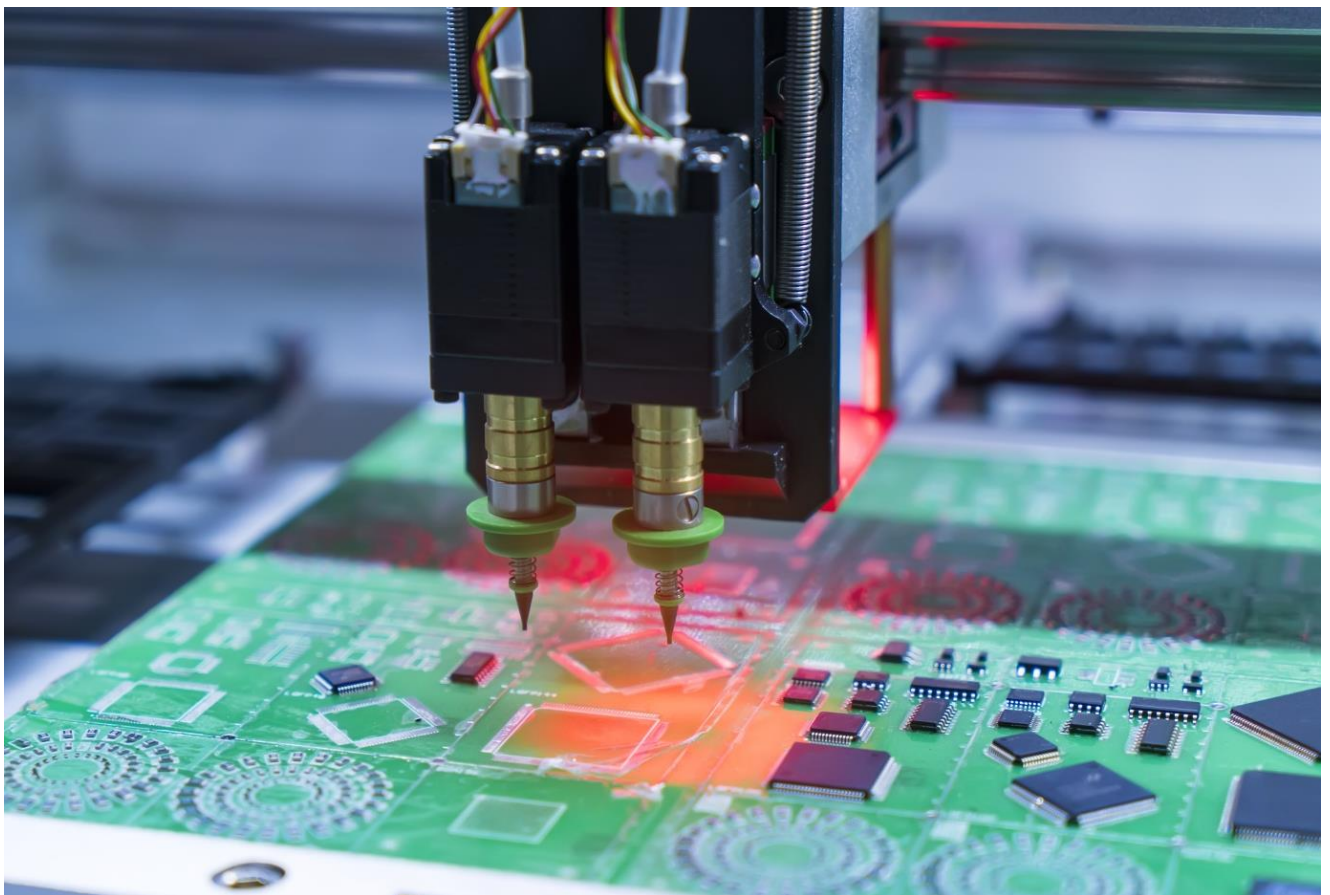


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Automated Manufacturing for Industrial Requirements

The advancing miniaturization of electronics components leads to a continuous increase in demands for high-efficiency production processes. Very dynamic production steps ensure efficient manufacturing conditions for contacting and mounting printed circuit boards or for a fully automated production, e.g. of smartphones. Precise dosing of highly viscous media, such as adhesives or soldering pastes, offers one possibility to increase the throughput of such production lines while fulfilling the quality requirements of miniaturized components. Piezo-based drive solutions are ideal here: Piezo actuators generate extreme dynamic movements and at the same time high forces. They are thus particularly suited for contactless dispensing of extremely fine liquid droplets, which can be generated with reproducible quality and at the highest speed.

Ultrafast Piezo Technology

Piezo actuators work with short response times and they move dynamically with displacements in the subnanometer range and frequencies of up to several thousand hertz. Furthermore, piezo actuators are maintenance-free, as they have no moving parts in the conventional sense. The motion of the actuators is based solely on crystalline solid-state effects (piezo effect), so there are no rotating or friction-generating mechanical components. Piezo elements also impress in terms of energy consumption, because they do not require any electrical power when at rest. They can therefore be used in a wide range of applications: In process technology e.g., short dosing cycles can be realized. Due to variable travel ranges, the dosing procedures can be finely adjusted and precisely controlled.

Contactless Droplet Generation with PipeJet Dispensers

The piezo-driven direct displacement method used by PipeJet dispensers is particularly interesting since it differs in crucial points from most common piezo dosing methods. The fluid line is made, for example, from an elastic polymer tube with a well-defined internal diameter. The actuator used extends along its longitudinal axis and can constrict the tube with a piston up to 100 times more than e.g. ring actuators (fig. 1). Therefore, it can provide sufficient force for the reliable dosing of highly viscous media and even fluids that contain particles, such as paints or nanoparticle suspensions, can easily be dosed in a precise droplet form.

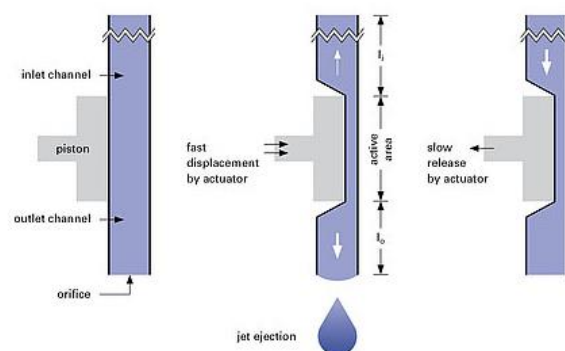


FIGURE 1: JET DISPENSING PRINCIPLE WITH PIEZO ACTUATORS.

Highly Dynamic Application of PICMA® Multilayer Actuators

PICMA® multilayer piezo actuators are particularly well suited for the application in high-precision dynamic dosing systems. The fast response behavior is a characteristic feature of piezo actuators. A change to the operating voltage brings about a very rapid displacement of the actuator. When the control voltage suddenly increases, a piezo actuator can reach its nominal

displacement in approximately one third of the period of its resonant frequency f_0 .

$$T \approx \frac{1}{3f_0}$$

With suitable drive electronics, piezo actuators can generate high accelerations of several ten thousand m/s^2 . As a result of the inertia of possible coupled masses as well as of the actuators themselves, dynamic pull forces occur that have to be compensated for with mechanical preloads. In sinusoidal operation, the maximum forces can be estimated as follows:

$$F_{\text{dyn}} \approx \pm 4\pi^2 m'_{\text{eff}} \frac{\Delta L}{2} f^2$$

With the F_{dyn} maximum dynamic pull force, the m'_{eff} effective mass of the piezo actuator, the ΔL displacement and the control frequency. Due to the layered structure made of thin piezo ceramic films of just a few micrometers in thickness, PICMA® actuators require comparatively low driving voltages of up to 120 V. The self-heating of the actuator should be taken into account when the displacement is at its fastest, e.g. during the dynamic droplet generation in jet dispensers.

Losses and Self-Heating

The maximum repetition frequency of piezoelectric actuators, when dynamically stressed, can be influenced by the resonant frequency of the mechanical system, the self-heating of the actuator, by limiting the electrical power of the control electronics, and – in closed-loop operation – it can be limited by the controller bandwidth. In open-loop applications and when the control takes place in the nominal voltage of the component, self-heating of the actuator usually limits the repetition frequency to several hundred hertz.

Dielectric losses in the actuator, when being operated below the first mechanical resonant frequency, are the main cause for self-heating. Due to the number of reversible ferroelectric domain reorientation, actuators have dielectric curves that are affected by hysteresis when higher control field strengths are applied. To load the actuator capacitance, the dielectric reactive energy flows into the actuator (fig. 2a). During unloading, a large part of this reactive energy flows back into the connected amplifier (fig. 2b). The content of the hysteresis area is the amount of energy that is converted into heat per cycle (fig. 2c).

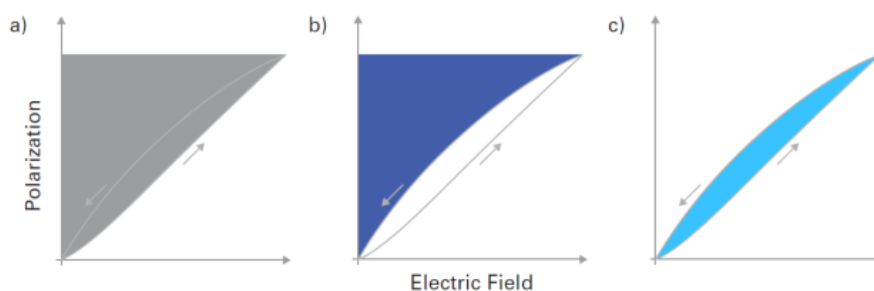


FIGURE 2: CHARACTERISTIC DIELECTRIC CURVES FOR ACTUATORS DURING UNIPOLAR OPERATION AND SCHEMATIC REPRESENTATION OF THE ENERGY FLOW: A) DIELECTRIC REACTIVE ENERGY DENSITY (DRED) FLOWING FROM THE AMPLIFIER INTO THE ACTUATOR DURING LOADING (GREY); B) DRED FLOWING BACK INTO THE AMPLIFIER WHEN THE ACTUATOR IS UNLOADED (BLUE); AND C) ENERGY LOSS DENSITY TRANSFORMED INTO HEAT IN THE ACTUATOR (LIGHT BLUE).

To describe the amount of dielectric losses in capacitive components, the dielectric loss factor $\tan\delta$ is generally used as the ratio of the resulting active thermal power P_w and the reactive power P_b . In addition, the loss factor strongly depends on the controlling field strength for different materials and drive modes of piezoceramic actuators. In unipolar operation at nominal field strength or voltage, approx. 10-15% of the reactive power used is transformed into heat output, depending on the material. In order to extend the application range of actuators to higher repetition rates, the generated heat must therefore be efficiently removed from the actuator.

Tests have shown that in PICMA® stack actuators of (W/D/H) 5x5x36 mm³ in size, with free convection of the surrounding air, only approx. 20% of the heat quantity is dissipated via the end surfaces. The remaining heat dissipates via the lateral surfaces. Since the actuators are enclosed in a very tight housing, free convection does not usually occur due to the limited amount of air.

Therefore, the use of electrically insulating heat conduction pastes or forced cooling with compressed air has become established as cooling measures for the lateral surfaces.

The impact (table 1) of different cooling measures, control signals types, and control voltage amplitudes was tested with non-encapsulated PICMA® stack actuators of (W/D/H) 5x5x36 mm³ in size. For the respective conditions described in table 1, the control frequency was gradually increased until the maximum temperature of 150°C was reached on the actuator surface. This temperature must be considered the maximum for using standard PICMA® actuators. Above 150°C, the soldered connection melts away. For applications above this temperature and up to 200°C, there are special solutions.

Signal	Voltage	Frequency reached 150°C [Hz] (actuator surface, central)			
		With housing	Clear convection	Air cooling [3.5 l/min]	Air cooling [25.5 l/min]
Sine	0...150 V	235	350	485	900 (140°C)
	0...120 V	355	535	750	
	0...100 V	510	765	1020	
	-15...105 V	300	428	560	
Rectangular	0...150 V	100	154	190	460
	0...120 V	125	193	240	745
	0...100 V	160	250	317	1060
	-15...105 V	130	201	265	735

TABLE 1: MAXIMUM DRIVING FREQUENCIES REACHED AT A SURFACE TEMPERATURE OF 150°C AT NON-ENCAPSULATED PICMA® STACK ACTUATORS (B/T/H) 5x5x36 MM³ WITH DIFFERENT COOLING MEASURES, TYPES OF DRIVING SIGNALS, AND DRIVING VOLTAGE AMPLITUDES; CONTROL WITH THE E-618 AMPLIFIER.

Heat Dissipation via the Actuator

The heat dissipation of piezoelectric multilayer actuators not only depends on the type of active cooling measures applied, but also on the geometric shape of the actuator and the surface available for radiation of the heat. The bigger the actuator, the smaller the respective ratio of actuator perimeter and cross-sectional area; the generated heat is then dissipated more poorly. With PICMA® actuators, 5x5x18 mm in size, it can be assumed that a power loss density of 1 W/cm³ by the actuator, which corresponds to about 10 to 20 K self-heating, is possible without significant performance losses. A power loss density of 10 W/cm³ can easily be dissipated with the help of active cooling for actuators with a cross-section smaller than 3x3 mm. The power loss densities of various PICMA® sizes are summarized in table 2.

Actuator Dimensions	PICMA		
	3 2	5 5	10 10
Circumference [mm]	10	20	40
Cross section [mm ²]	6	25	100
Relation (Circumference/Cross Section)	1.67	0.8	0.4
Equivalent loss power density limit (no cooling) [W/cm ³]	2.1	1.0	1.5

TABLE 2: POWER LOSS DENSITY OF DIFFERENT PICMA® ACTUATORS WHEN SELF-HEATING DURING DYNAMIC OPERATION.

In table 3, the thermal ambient factors for efficient heat dissipation is shown using PICMA® actuators 5x5x36 mm in size as an example. It shows that encapsulated actuators have the lowest level of heat dissipation. Active cooling

with oil brings about the most effective heat dissipation in the actuators.

Actuator type and cooling	Thermal ambient factors
Encapsulated PICMA® actuators	1
Free convection (non-encapsulated actuators)	1.5
Active air cooling (low flow rates)	2
Heat-conducting paste	5
Active air cooling (high flow rates)	5
Active cooling in oil	8-10

TABLE 3: THERMAL AMBIENT FACTORS FOR HEAT DISSIPATION SHOWN USING PICMA® ACTUATORS AS AN EXAMPLE: 5x5x36 MM IN SIZE AT 150°C.

Which Actuators Are Suitable?

Fig. 3 shows that active air cooling allows repetition frequencies about 4 to 5 times higher than those of the housed state without free convection. An additional increase of up to factor 8 to 10 can be achieved by cooling the encapsulated PICMA® stack actuators with liquid. While air cooling allows repetition frequencies up to a maximum of 1,000 Hz, a frequency of 3,300 Hz can be proven with liquid cooling at 100 V_{pp} or 35 µm induced displacement; in addition, at about 80°C self-heating, a still significant difference to the maximum temperature of 150°C can be maintained. The upper frequency limit in the test is determined by the small signal bandwidth of the E-617KDYN amplifier.

With high repetition frequencies, very high cycle times can be achieved within a very short time by the actuators. In lifetime tests, it was proven that PICMA® stack actuators, (W/D/H) 5x5x36 mm³ in size at 0...100 V sine, can reach 10¹⁰ cycles without

any problems. This cycle number was proven on five samples each of non-encapsulated and encapsulated actuators with heat dissipation. Thanks to the combination of encapsulated PICMA® stack actuators with heat dissipation, active liquid cooling, and the E-617KDYN amplifier, a continuous operation with 35 µm displacement at 3,300 Hz sine control was demonstrated in the studies.

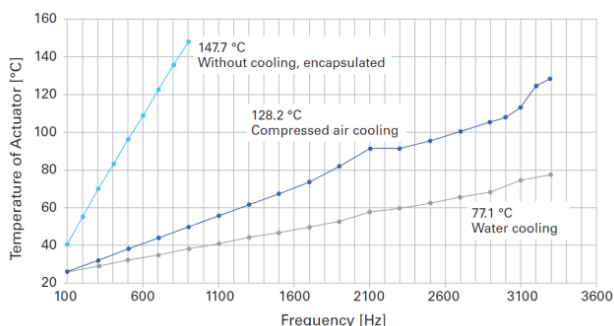


FIGURE 3: SELF-HEATING TEST OF ENCAPSULATED PICMA® STACK ACTUATORS (W/D/H) 5X5X36 MM³ IN SIZE WITH HEAT DISSIPATION USING DIFFERENT COOLING MEASURES; CONTROL WITH 0... 100 V SINE; E-617KDYN AMPLIFIER.

High-Current PICMA® Multilayer Actuators

For the dynamic displacement of PICMA® actuators, e.g. in dosing applications, not just the self-heating of the actuators caused by the sudden high frequency displacement plays an important role. The current capacity and therefore the temperature stability of the electrical contacts are also important design parameters. During high-dynamics operation of the multilayer actuators, extremely fast voltage signals with triggering edges of 100-200 µs are used. In order to effectively load and unload the capacity of a multilayer actuator in high dynamics, a lot of current has to flow in a very short time. This strong current flow leads to a local temperature increase in the contacting of the actuators that can even lead to the soldering points melting in

the actuator contacting. PICMA® piezo multilayer actuators with high-current electrodes are very well suited for these dynamic application areas (fig. 4). The meander-shaped contact rail is connected to the actuator with two large soldering tracks. With this, peak currents of up to 20 A can be reached when controlling the actuator. Therefore, the electrical contacting of the actuators makes fastest movements possible; due to their structure it adapts very well to the pull forces and displacements of the actuators.

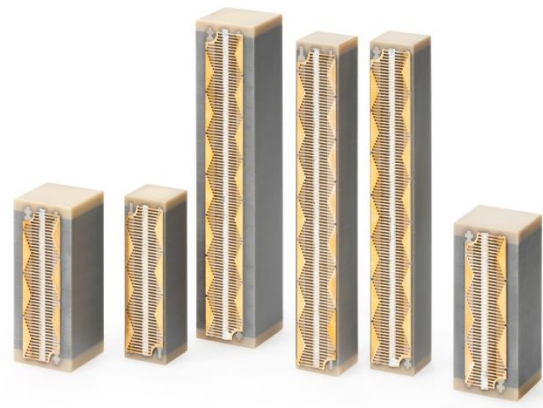


FIGURE 4: PICMA® PIEZO MULTILAYER ACTUATORS WITH HIGH-CURRENT ELECTRODES FOR DYNAMIC OPERATION.

Conclusion

PICMA® piezo multilayer actuators are ideally suited for use in high-dynamics dispensing applications. With active cooling, highest repetition frequencies of 3,300 Hz can be achieved – with a suitable electronics even more. The high-current design of the electrical contacting ensures a reliable functioning of the actuators, even at peak currents of up to 20 A.

About the Author



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PI Ceramic in Brief

Redefining the limits of what can be measured and moved, together with our customers: As a worldwide partner with more than 25 years of expertise, PI Ceramic develops and manufactures sophisticated piezoceramic components and subsystems in the areas of industrial ultrasonic sensors, precision dosing, optics, and semiconductor or medical technology. Seventy of the currently 320 employees at the location in Thuringia, Germany, work in research and development. PI Ceramic is part of the PI Group, the innovation and market leader for high-precision positioning technology.